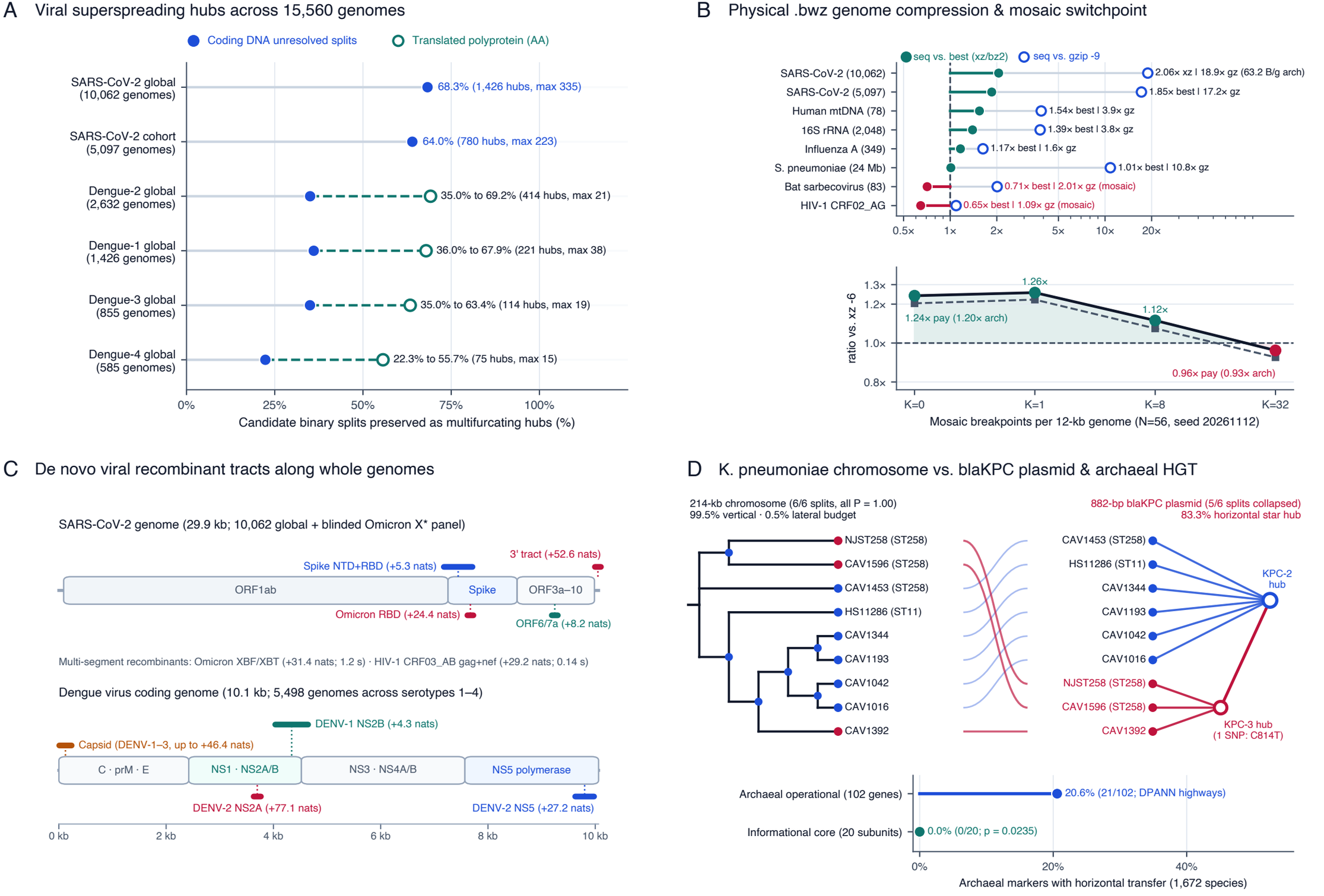
A
Viral superspreading hubs across 15,560 genomes
Coding DNA unresolved splits
Translated polyprotein (AA)
SARS-CoV-2 global (10,062 genomes)
68.3% (1,426 hubs, max 335)
SARS-CoV-2 cohort (5,097 genomes)
64.0% (780 hubs, max 223)
Dengue-2 global (2,632 genomes)
35.0% to 69.2% (414 hubs, max 21)
Dengue-1 global (1,426 genomes)
36.0% to 67.9% (221 hubs, max 38)
Dengue-3 global (855 genomes)
35.0% to 63.4% (114 hubs, max 19)
Dengue-4 global (585 genomes)
22.3% to 55.7% (75 hubs, max 15)
0%
25%
50%
75%
100%
Candidate binary splits preserved as multifurcating hubs (%)
B
Physical .bwz genome compression & mosaic switchpoint
seq vs. best (xz/bz2)
seq vs. gzip -9
SARS-CoV-2 (10,062)
2.06× xz | 18.9× gz (63.2 B/g arch)
SARS-CoV-2 (5,097)
1.85× best | 17.2× gz
Human mtDNA (78)
1.54× best | 3.9× gz
16S rRNA (2,048)
1.39× best | 3.8× gz
Influenza A (349)
1.17× best | 1.6× gz
S. pneumoniae (24 Mb)
1.01× best | 10.8× gz
Bat sarbecovirus (83)
0.71× best | 2.01× gz (mosaic)
HIV-1 CRF02_AG
0.65× best | 1.09× gz (mosaic)
0.5×
1×
2×
5×
10×
20×
ratio vs. xz -6
1.3×
1.2×
1.0×
0.8×
1.26×
1.12×
1.24× pay (1.20× arch)
0.96× pay (0.93× arch)
K=0
K=1
K=8
K=32
Mosaic breakpoints per 12-kb genome (N=56, seed 20261112)
C
De novo viral recombinant tracts along whole genomes
SARS-CoV-2 genome (29.9 kb; 10,062 global + blinded Omicron X* panel)
3' tract (+52.6 nats)
Spike NTD+RBD (+5.3 nats)
ORF1ab
Spike
ORF3a–10
Omicron RBD (+24.4 nats)
ORF6/7a (+8.2 nats)
Multi-segment recombinants: Omicron XBF/XBT (+31.4 nats; 1.2 s) · HIV-1 CRF03_AB gag+nef (+29.2 nats; 0.14 s)
Dengue virus coding genome (10.1 kb; 5,498 genomes across serotypes 1–4)
DENV-1 NS2B (+4.3 nats)
Capsid (DENV-1–3, up to +46.4 nats)
C · prM · E
NS1 · NS2A/B
NS3 · NS4A/B
NS5 polymerase
DENV-2 NS2A (+77.1 nats)
DENV-2 NS5 (+27.2 nats)
0 kb
2 kb
4 kb
6 kb
8 kb
10 kb
D
K. pneumoniae chromosome vs. blaKPC plasmid & archaeal HGT
214-kb chromosome (6/6 splits, all P = 1.00)
99.5% vertical · 0.5% lateral budget
882-bp blaKPC plasmid (5/6 splits collapsed)
83.3% horizontal star hub
NJST258 (ST258)
CAV1596 (ST258)
CAV1453 (ST258)
HS11286 (ST11)
CAV1344
CAV1193
CAV1042
CAV1016
CAV1392
CAV1453 (ST258)
HS11286 (ST11)
CAV1344
CAV1193
CAV1042
CAV1016
NJST258 (ST258)
CAV1596 (ST258)
CAV1392
KPC-2 hub
KPC-3 hub (1 SNP: C814T)
Archaeal operational (102 genes)
20.6% (21/102; DPANN highways)
Informational core (20 subunits)
0.0% (0/20; p = 0.0235)
0%
20%
40%
Archaeal markers with horizontal transfer (1,672 species)